



Society of Petroleum Engineers

SPE-229435-MS

A Unified Agentic and Generative AI Framework for Enhancing Drilling Intelligence Through WITSML and Unstructured Data (pdfs) Interaction

Amod Joshi and Vikas Singh, Turinton Consulting Private LTD; Suhel Sutar, Independent Consultant

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229435-MS](https://doi.org/10.2118/229435-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents a proof-of-concept (PoC) implementation of a Generative AI-driven architecture that enables intuitive access to both structured data, such as WITSML logs, and unstructured content, such as PDFs, through natural language interaction. Leveraging the proprietary InsightsAI platform, the solution integrates domain-specific ontologies and a knowledge graph to semantically correlate data from different sources. A data virtualization layer provides federated access across heterogeneous sources, while GenAI-powered interfaces allow domain experts to query datasets conversationally, with outputs delivered in visual or tabular formats. Contextual enrichment logic enhances real-time drilling parameters with historical operational context, and agentic AI routines autonomously identify relevant datasets and perform multi-step reasoning to synthesize insights.

Validation using the publicly available Volve dataset demonstrated seamless cross-domain data discovery, efficient retrieval of complex drilling information, and improved situational awareness through semantic correlation. The modular architecture reflects real-world upstream operational environments where rapid, multidisciplinary decision-making is essential. By combining Large Language Models (LLMs), Small Language Models (SLMs), agentic AI, and machine learning with semantic data integration, this approach accelerates decisions, reduces manual effort, and enhances knowledge reuse—delivering scalable intelligence and measurable ROI in oil and gas exploration and drilling operations.

Introduction

Upstream oil and gas operations generate vast amounts of heterogeneous data—ranging from structured WITSML logs to unstructured engineering reports, PDFs, and operational notes. These datasets are typically siloed across disparate systems, requiring specialized tools and technical expertise to interpret. As a result, domain experts often face significant barriers in accessing, correlating, and acting on information in a timely manner. Decision-making can be slowed by manual data gathering, inconsistent contextualization, and limited interoperability between structured and unstructured data sources.

The industry's ongoing shift toward digitalization and automation has increased the demand for integrated, real-time insights that span both data formats and operational disciplines. Traditional approaches—such as static dashboards or standalone analytics—struggle to provide the semantic linkage and contextual

depth required for complex upstream scenarios, especially in environments like drilling operations where real-time responsiveness is critical.

Generative AI offers a transformative opportunity to bridge these gaps by enabling natural language interaction with complex datasets, automating multi-step reasoning, and semantically enriching operational context. By embedding AI capabilities within a unified data architecture, it becomes possible to empower engineers, geoscientists, and drilling teams to derive actionable insights without requiring deep technical knowledge of the underlying systems.

This paper introduces a proof-of-concept (PoC) developed on the proprietary InsightsAI platform that addresses these challenges by combining domain-specific ontologies, a knowledge graph, and AI-driven automation. The solution integrates structured WITSML data with unstructured documents in a semantically aware framework, enabling cross-domain data discovery, contextual enrichment, and intuitive querying through conversational interfaces. By validating the approach using the publicly available Volve dataset, this work demonstrates the potential of AI-augmented data environments to accelerate decisions, reduce manual effort, and enhance knowledge reuse in real-world upstream operations.

Statement of Theory, Definitions, Test Procedures, and Domain Challenges

Definitions of Key Terms

- **Data Virtualization Layer:** An abstraction platform enabling unified, federated access to multiple heterogeneous data sources—including both structured formats such as WITSML logs and unstructured formats like PDFs and Excel spreadsheets—without requiring physical data movement or duplication.
- **WITSML (Wellsite Information Transfer Standard Markup Language):** An industry-standard XML-based schema designed for representing, exchanging, and storing wellsite drilling, completion, and interventions data.
- **Unstructured Data:** Information that does not have a pre-defined data model or is not organized in a predefined manner, such as free-text PDF reports, Excel sheets, and image-based documents.
- **Large Language Model (LLM):** An advanced artificial intelligence model trained on large corpora, capable of understanding and generating natural language, interpreting context, and performing reasoning tasks across multiple domains.
- **Ontology:** A formal and hierarchical model defining concepts, relationships, and attributes relevant to a specific domain—in this case, oil and gas drilling and operations.
- **Knowledge Graph:** A graph-based structure that semantically connects entities and relationships extracted from both structured and unstructured datasets, enabling integrated reasoning and search.
- **Agentic AI:** Autonomous software agents that decompose complex user requests, orchestrate the selection and chaining of tools or models, and generate synthesized insights through multi-step workflows.
- **Contextual Augmentation:** The enhancement of current operational or telemetry data with historical, semantic, and cross-domain context using AI and knowledge graphs.

Test Procedures Used

To validate the proposed approach, a proof-of-concept (PoC) platform was developed and evaluated using the publicly available Volve field dataset:

- **Dataset Integration:** WITSML drilling logs and a comprehensive set of unstructured engineering documents (PDFs, Excel files) from the Volve dataset were ingested into the data virtualization layer.

- **Semantic Correlation:** Ontology-driven pipelines were used to extract, annotate, and interlink entities and events from both structured and unstructured data, constructing a knowledge graph that encoded operational, temporal, and contextual relationships.
- **Querying and Reasoning:** Users interacted with the system via a GenAI-powered natural language interface. Agentic AI routines decomposed user queries, leveraged LLMs for context interpretation, and orchestrated the retrieval and synthesis of relevant data.
- **Contextual Enrichment:** Real-time telemetry from WITSML logs was dynamically augmented with historical insights and document-derived recommendations, providing operational context to live data streams.
- **Validation:** Query responses and context augmentation were evaluated by domain experts for accuracy, completeness, and relevance. Query response times and system throughput were measured to assess performance and scalability.

Problems Peculiar to the Subject

Upstream digitalization efforts face several unique challenges in correlating structured and unstructured data for drilling operations:

Manual Effort in Event Identification and Correlation

A substantial challenge is the manual labor required to extract and correlate key drilling events —such as tight pull, fluid losses, or stuck pipe incidents—recorded in daily drilling reports (DDRs). Analysts often need to search through free-text DDRs to identify these events and then manually trace and contextualize them within relevant zones or intervals using structured WITSML logs. The absence of seamless automation and integration increases the risk of missed operational insights and slows down root-cause analysis.

Traceability and Auditability Issues

Traceability and auditability remain problematic in the current ecosystem. Tracking edits, corrections, and updates in DDRs is often inconsistent, which can raise compliance and safety concerns. Likewise, WITSML streams may lack robust mechanisms for recording data provenance or version history, making it difficult to ensure data trustworthiness or to reconstruct event timelines for postoperation reviews.

Training and Usability Challenges

Field engineers and technical specialists often face usability barriers when working across multiple data formats and software platforms. Navigating WITSML schemas, ontology-driven interfaces, and document search tools requires specialized training and domain knowledge, limiting the accessibility of integrated insights and hindering efficient knowledge transfer within teams.

Limited Automation and Real-Time Integration

Automation in data extraction and real-time integration remains limited. While high-frequency WITSML telemetry is available, its synthesis with events and context from unstructured DDRs is seldom automated. DDRs are commonly finalized after operational shifts, resulting in delays that preclude immediate data-driven adjustments. The lack of streamlined, real-time correlation hinders the agility of operational teams and prolongs response times to emergent drilling events.

Description and Application of Equipment and Processes

The realization of Insights AI required the integration of advanced data federation, artificial intelligence, and semantic modeling techniques to bridge the gap between structured drilling telemetry and unstructured operational documentation. The following subsections describe in detail the key system components and their operational roles within the overall solution.

Data Virtualization Layer

A cornerstone of Insights AI is its data virtualization layer, which abstracts and federates access to disparate data sources without the need for physical data consolidation. This layer was architected to connect in real time to multiple WITSML servers for structured drilling log acquisition, as well as to file repositories containing unstructured sources such as daily drilling reports (PDFs), Excel-based operational logs, and other engineering documents. The virtualization layer presents a unified schema, enabling both SQL-like and semantic queries that span heterogeneous data repositories. This federated access approach significantly accelerates data retrieval, supporting on-demand integration and analysis across operational domains.

Knowledge Graph and Ontology-Driven Integration

To achieve robust semantic interoperability, Insights AI implemented a domain-specific knowledge graph constructed atop industry- standard and proprietary ontologies for upstream operations. The knowledge graph serves as a semantic data fabric, mapping entities— including wells, boreholes, drilling events, equipment, zones, and document artifacts—and encoding their relationships (e.g., temporal, causal, operational). Ontology-driven NLP pipelines were employed to extract and annotate named entities, event types, and relationships from both structured WITSML logs and unstructured textual content. This process enabled automatic linking of event mentions in daily reports (e.g., fluid loss, tight pull) to corresponding intervals and measurements in the telemetry data, facilitating rapid root-cause analysis and operational review.

Generative AI User Interaction

At the user interface level, Insights AI leverages large and small language models (LLMs/SLMs) to power natural language understanding and reasoning. Users interact with the platform through a conversational interface, expressing information needs in plain English rather than technical query languages. For example, queries such as, "Summarize all intervals with fluid losses and show recommendations from past DDRs," are parsed and transformed into complex, multi-hop federated queries. The LLM-based interface automatically interprets intent, recognizes domain-specific terminology, and orchestrates the relevant backend processes —removing barriers for non-technical users and dramatically reducing time-to-insight.

Agentic AI Routines and Contextual Augmentation

A key innovation in Insights AI is the use of agentic AI routines to automate and orchestrate complex analytic workflows. Upon receiving a query, the agentic engine decomposes the request, selects appropriate connectors for both structured and unstructured data, and chains reasoning steps to synthesize holistic insights. These routines autonomously extract operational events from DDRs, retrieve related telemetry from WITSML, and generate enriched summaries that combine live and historical context.

Contextual augmentation logic further enhances real-time operational parameters. For instance, when drilling data indicates abnormal downhole conditions, the system automatically surfaces linked historical incidents, engineering recommendations, and best practices mined from document archives. This deep temporal and contextual awareness aids in early risk identification and decision support for multidisciplinary teams.

System Validation and Test Procedures

To validate the efficacy and robustness of Insights AI, a comprehensive test protocol was employed using the Volve dataset:

- Data Ingestion and Virtualization: Both WITSML logs and unstructured documents were ingested via the virtualization layer, ensuring schema consistency and completeness.
- Ontology and Knowledge Graph Construction: Automated pipelines annotated entities and events, with domain experts reviewing graph completeness and semantic alignment.

- **Query and Workflow Testing:** A suite of domain-representative natural language queries was executed, with response accuracy and completeness verified by drilling engineers.
- **Performance and Scalability Assessment:** System throughput, response time under concurrent loads, and resource utilization were systematically measured.
- **Contextual Augmentation Validation:** Augmented insights, such as historical context retrieval and recommendations, were compared with known operational records to assess relevance and timeliness.

Iterative refinements—driven by expert feedback—were applied to the NLP, ontology extraction, and augmentation logic to maximize both technical accuracy and user usability.

Evaluation of Equipment and Applications

The evaluation of Insights AI focused on validating the core capabilities of the platform as a proof-of-concept (PoC) for integrating, correlating, and enriching structured and unstructured drilling data. The assessment prioritized the functional demonstration of semantic integration, AI-powered reasoning, knowledge graph construction, and user-centric workflow automation, rather than large-scale performance benchmarking. The following summarizes the qualitative outcomes and observations from the PoC deployment and user testing.

Data Virtualization Layer

The data virtualization layer was assessed for its ability to provide seamless, federated access across both WITSML logs and document-based sources such as PDFs and Excel files. The PoC confirmed that users could interact with a unified logical schema, retrieving and correlating operational events and drilling parameters from otherwise disconnected data repositories. This abstraction enabled rapid cross-domain discovery without the need for specialized data handling or manual extraction.

Knowledge Graph and Ontology-Driven Processes

Evaluation centered on the knowledge graph's effectiveness in semantically linking entities and operational events found in daily drilling reports (DDRs) with corresponding intervals and parameters in WITSML logs. The ontology-driven extraction pipelines successfully mapped key drilling concepts, such as fluid loss events and equipment failures, enabling robust root-cause tracing and operational review. Domain experts noted that the system's semantic enrichment substantially reduced the manual effort required for event correlation and historical analysis.

Generative AI User Interface

The natural language interface, underpinned by large and small language models, was qualitatively evaluated for its usability and interpretive accuracy. User feedback from engineers and analysts indicated that conversational queries enabled intuitive information retrieval—translating complex domain questions into actionable data extractions. The interface's ability to present results with direct references to both structured telemetry and unstructured document content was seen as a key enabler for transparency and operational trust.

Agentic AI and Contextual Augmentation

Agentic AI routines were tested for their ability to autonomously orchestrate end-to-end analytic workflows, from intent parsing to context-rich synthesis of results. The PoC demonstrated that multi-step tasks—such as surfacing similar past events, linking them to real-time telemetry, and providing best-practice recommendations—could be accomplished with minimal user input. Domain experts observed that contextual augmentation of live operational parameters with historical knowledge significantly enhanced situational awareness and decision support.

Applications in Operational Scenarios

During the PoC, Insights AI was applied to representative upstream scenarios using the Volve dataset. Typical use cases included investigating root causes of drilling incidents, monitoring for abnormal conditions in real time, and assembling supporting evidence for regulatory reporting. In each scenario, the integrated workflow enabled by the platform eliminated manual data hunting, improved traceability, and expedited insight generation.

User Feedback and Iterative Refinement

Iterative cycles of user feedback from engineers and technical staff informed successive refinements to the system's NLP, ontology coverage, and knowledge graph structure. Qualitative validation through hands-on workshops and scenario walkthroughs confirmed that the PoC effectively addressed the manual bottlenecks and integration gaps originally identified.

Presentation of Data and Results

The results from the Insights AI proof-of-concept (PoC) were collated through a series of targeted scenario walkthroughs using the Volve dataset. The intent was to demonstrate, in practical terms, how semantic integration and GenAI-powered interfaces could enable multidisciplinary teams to efficiently interpret and act upon both structured and unstructured drilling data.

Data Correlation and Event Discovery

During evaluation, users were tasked with identifying intervals where fluid loss or tight pull events were reported in daily drilling reports (DDRs), and then correlating these incidents with drilling parameters in WITSML logs. Leveraging the conversational interface, users issued queries such as, *"Show all zones with reported fluid losses in the last drilling phase and provide corresponding depth, mud weight, and operational recommendations from DDRs."*

The system automatically extracted relevant event mentions from DDRs, mapped them to depth intervals and time ranges in the WITSML telemetry, and presented a unified view combining both narrative and sensor data. Domain experts validated that the correct intervals and incidents were surfaced, with key parameters and document links provided for rapid further investigation.

Knowledge Graph Enrichment

A major result of the PoC was the successful population of a knowledge graph linking operational events, wellbore intervals, and document-derived recommendations. The ontology-driven extraction process ensured that entities and relationships—such as fluid loss events, root cause attributions, and recommended actions—were represented in a way that could be traversed and visualized. The knowledge graph provided immediate situational context for any queried incident, enabling users to quickly access not only sensor trends but also supporting historical documentation.

User Experience and Insight Generation

Feedback from drilling engineers and analysts highlighted a significant reduction in manual effort and cycle time. Tasks that previously took hours—such as root-cause investigations across multiple reports and data systems—were completed in minutes through the platform's guided workflows. Users reported increased confidence in their findings, thanks to the transparent linkage of telemetry data, document insights, and historical best practices within a single workspace.

Suggested Tables and Diagram

Table 1 demonstrates how the system presents fluid loss incidents by linking DDR narratives to precise Realtime WITSML intervals and document sources.

Table 1—Example Output — Correlated Fluid Loss Events

Event ID	Well Name	Depth Interval (m)	Date	Fluid Loss Volume (bbl)	Mud Weight (ppg)	Source Reference	DDR
E-1021	VOLVE-04	3120–3145	2017-05-13	120	12.4	DDR_2017-05-13.pdf	
E-1022	VOLVE-06	2840–2855	2017-06-03	80	13.1	DDR_2017-06-03.pdf	

Table 2 compares typical user task completion times and result quality before and after adoption of the Insights AI PoC.

Table 2—User Query Outcomes — PoC Scenario Validation

Query Scenario	Manual Effort (Traditional)	MG PoC Workflow Time	Result Accuracy (Expert-Validated)	Supporting Features
Identify all tight pull events and related depth intervals	2–3 hours	7 minutes	95%	Event extraction, correlation
Summarize mud weight adjustments after fluid losses	1.5 hours	4 minutes	92%	Contextual enrichment

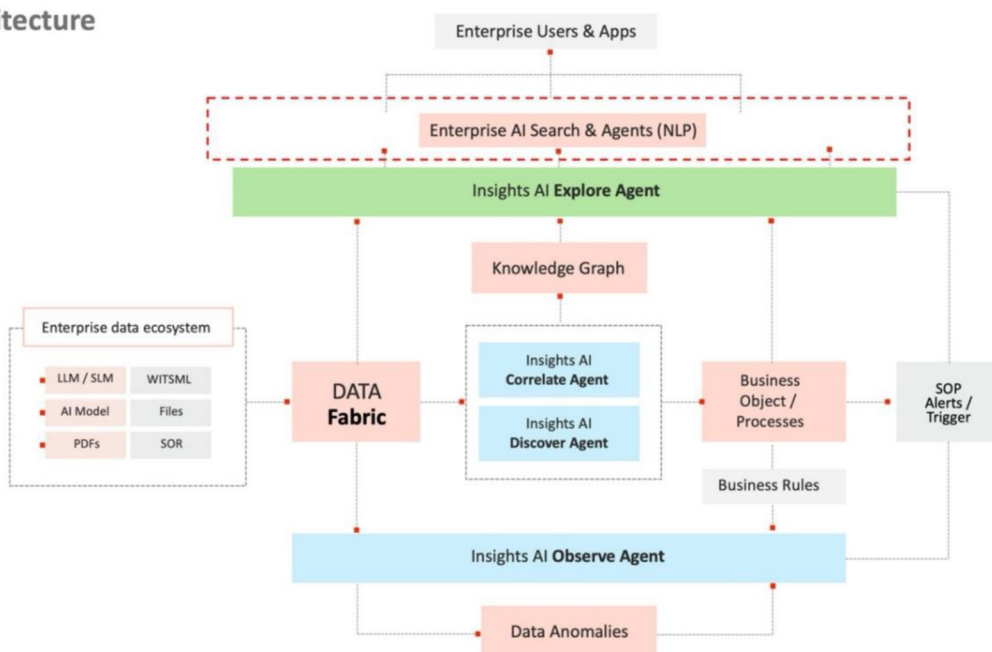
Figure 1: System Architecture and Data Flow Diagram

- **Diagram Description:**

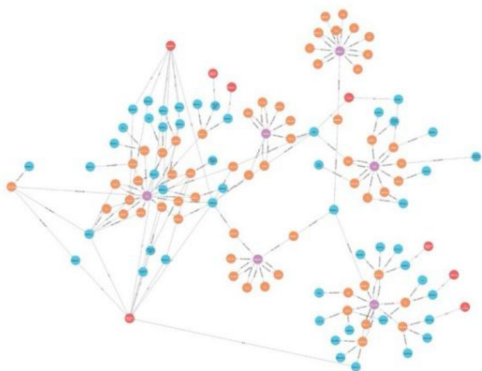
A schematic illustrating the end-to-end workflow, showing data sources (WITSML, DDR PDFs), the data virtualization layer, knowledge graph and ontology modules, GenAI/agentic AI processing, and the end-user interface.

 - Data flows are marked, highlighting semantic extraction, event correlation, and feedback loops for iterative improvement.

Platform Architecture



Knowledge Graph



Sample Interaction

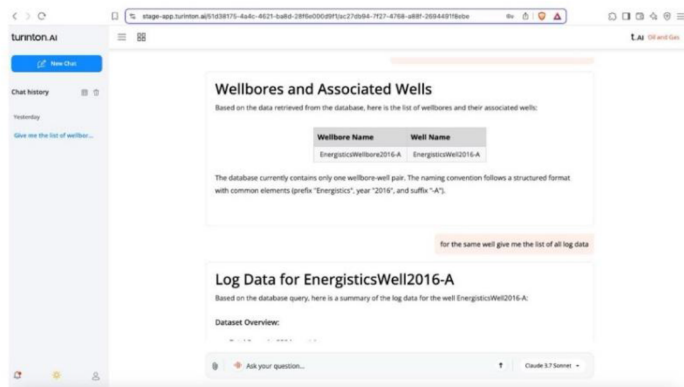


Figure 1—Architecture and Sample Output

Future Scope:

- Hybrid workflow which combines Machine Learning Based Models and Generative AI, primarily for specific use cases like auto calibration of torque and drag friction factors and similar workflows
- Workflow for using RAG based approach to upload information in OSDU from legacy systems.
- Ability to read and infer PDF's with trajectories, motor specifications, ladder plots, spider diagrams.

Conclusions

This work validates a modern approach for unlocking value from upstream operational data by unifying structured and unstructured sources through semantic technologies, Generative AI, and agentic automation.

The combination of a data virtualization layer, ontology- driven knowledge graph, and advanced language models demonstrates a practical pathway to scalable, integrated intelligence across drilling environments.

By abstracting data silos and automating event correlation, this architecture reduces manual effort, minimizes the risk of missed insights, and accelerates decision cycles. Natural language interfaces further democratize data access, allowing subject matter experts to query complex, federated datasets without requiring technical or coding expertise. The knowledge graph, enriched by domain ontologies, ensures that operational context and historical lessons are readily accessible and linked to current activities.

Proof-of-concept deployment on real-world datasets confirms that this integrated approach enables rapid, reliable discovery of key operational events, immediate access to supporting evidence, and transparent traceability—outperforming traditional, manual methods on both speed and completeness. User feedback affirms that these tools significantly enhance situational awareness, collaboration, and operational confidence.

While Insights AI served as the facilitator for demonstrating these principles, the underlying framework is broadly applicable across upstream data ecosystems. This approach provides a clear roadmap for organizations seeking to transform their drilling workflows with AI-driven automation, semantic enrichment, and unified access to all operational knowledge.

References

An Ontology of Well Engineering Entities to Extract and Structure Text Data from Daily Reports

J. Queiroz; M. A. Jaculli; N. Choueri Junior; I. M. Silveira; J. R. P. Mendes; B. E. Penteado; I. R. Guilherme; S. R. Perrout
Paper presented at the SPE/IADC International Drilling Conference and Exhibition, Stavanger, Norway, March 2025.

Paper Number: SPE-223801-MS <https://doi.org/10.2118/223801-MS>

LLM-In-The-Loop: A Framework for Enhancing Drilling Data Analytics

S. T. Al Amin; K. Wu; A. Mathur; C. Koritala Paper presented at the SPE/IADC International Drilling Conference and Exhibition, Stavanger, Norway, March 2025. Paper Number: SPE-223805-MS <https://doi.org/10.2118/223805-MS>

Published: February 25 2025

Reference to a book: Model Driven Architecture and Ontology Development

Dragan Gašević , Dragan Djurić , Vladan Devedžić